

MATH 2250: DIFFERENTIAL EQUATIONS AND LINEAR ALGEBRA

Section 002

Summer 2023

Instructor:	John Turnage	Time:	Weekdays 10:45 – 11:45am
Email:	turnage@math.utah.edu	Place:	LCB 219

General Information

Course Page: All official announcements, problem sets, quizzes, deadlines, and grades will be posted online via the course Canvas page. Homework assignments and labs are to be submitted to the course Gradescope page, and weekly quizzes will be given via Canvas.

Office Hours: Office hours will be held on **Tuesdays and Thursdays** from **12-1 pm** in the **LCB loft**. The loft is on the fourth floor of the LCB and may be accessed via the stairwells on the east side of the building or via an elevator on the third floor.

Textbook: The following textbook is required:

Differential Equations and Linear Algebra, Edwards (ISBN: 9780134497181).

An e-textbook version is available through the university's inclusive access program and will be accessible through the 'bookshelf' tab on the course canvas page. Students should receive an email a week prior to the start of class allowing them to opt out of the e-textbook rental. If students do nothing, they will automatically be opted-in and charged the rental fee.

Important Dates:

A more complete course schedule is provided on page three.

Classes Begin	Monday, May 15
Drop Deadline	Wednesday, May 24
Memorial Day Holiday	Monday, May 29
Midterm #1	Friday, June 9
Juneteenth Holiday	Monday, June 19
Withdrawal Deadline	Friday, June 23
Midterm #2	Friday, June 30
Independence Day Holiday	Tuesday, July 4
Midterm #3	Friday, July 21
Pioneer Day Holiday	Monday, July 24
Classes End	Wednesday, August 2
Final Exam	Thursday, August 3 from 10 am – 12 pm

Syllabus Subject to Change: This syllabus is meant to serve as an outline for the course goals and objectives. Please note that I may modify it with reasonable notice to you.

Email Policy: I do not typically check emails during the hours of 6pm and 8am on weekdays, and I rarely check them at all on the weekends. Messages received during these times will receive attention once I am back online. Otherwise, you can expect to receive a response to your message within 6 hours, often much sooner.

Coursework and Grading

Homework (20%): Every Monday, a homework assignment will be released covering material to be presented during the week's lectures and possibly covering material from the previous week's lectures. **Homework will be due at 11:59 pm on the following Tuesday, to be submitted via Gradescope.** There will be a total of 11 homework assignments during the semester: at least **two of a student's lowest homework scores will be dropped.** A portion of each assignment will be graded for completion, so please attempt every question.

Quizzes (10%): On most Fridays after class, a short 1-4 problem quiz will be given through Canvas, **due by Midnight**, and taking roughly 10-25 minutes to complete. Quizzes may be taken at a self-selected time, and you will have one hour to complete them. These problems will cover material presented in that week's lectures and are meant to be a straightforward check of your understanding. There will be 8 quizzes throughout the semester, and at least **two of a student's lowest quiz scores will be dropped.**

Lab (15%): Most Thursdays, a directed problem solving lab will be held. Students will be split into groups of 4-6 and given a few challenge problems. These problems are generally longer and more in-depth than those assigned for homework. The lab work serves to increase problem solving fluency by helping students to develop the requisite skills to solve problems requiring multiple coordinated skills, including: interpretation and identification of relevant variables and unknowns, operationalization of the question into a series of executable methods, and interpretation and communication of results. Such skills are invaluable for any scientist or engineer working in a group environment to solve complex problems. There will be 7 labs throughout the semester, and **a student's lowest lab score will be dropped.**

Midterm Exams (30%): There will be three midterms on select Fridays throughout the semester. Practice problems and a practice exam will be released about a week prior to the exams, and lab sections falling on a Thursday preceding an exam will be used for review. **A student's lowest midterm score will be dropped.**

Final Exam (25%): There will be a two-hour comprehensive final exam given at the University-designated time. As with the midterms, review material will be provided about one week prior to the exam, and we will hold at least one day of review in class.

Letter Grades: Letter grades will be determined as follows: If X is your final percentage grade, then

$X \geq 93\% \implies A$	$X \geq 90\% \implies A-$
$X \geq 87\% \implies B+$	$X \geq 83\% \implies B$
$X \geq 83\% \implies B$	$X \geq 80\% \implies B-$
$X \geq 77\% \implies C+$	$X \geq 73\% \implies C$
$X \geq 70\% \implies C-$	$X \geq 67\% \implies D+$
$X \geq 63\% \implies D$	$X \geq 60\% \implies D-$
$X < 60\% \implies E$	

Grade Adjustments: The final exam tests knowledge learned over the entire course, and later material is over-represented since it merges all of the prior topics into a coherent whole. If a student shows strong performance on the final exam, exceeding prior performance, their course grade may be adjusted upward to reflect the student's improvement. The specific rules for this grade adjustment will be devised by the instructor after the final exam is completed, and any adjustment will be applied evenly to all students.

Tentative Course Schedule

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1: May 15-19	Sec. 1.1 <i>Start homework 1</i>	Sec. 1.2	Sec. 1.3	Lab 1	Sec. 1.4 <i>Canvas quiz 1</i>
Week 2: May 22-26	Sec. 1.5 <i>Start homework 2</i>	Sec. 2.1 <i>Homework 1 due</i>	Sec. 2.2 Drop deadline	Lab 2 <i>Lab 1 due</i>	Sec. 2.3 <i>Canvas quiz 2</i>
Week 3: May 29- June 2	Memorial Day <i>Start homework 3</i>	Sec. 2.4 <i>Homework 2 due</i>	Sec 2.5	Lab 3 <i>Lab 2 due</i>	Sec. 3.1 <i>Canvas quiz 3</i>
Week 4: June 5-9	Sec 3.2 <i>Start homework 4</i>	Sec. 3.3 <i>Homework 3 due</i>	Sec. 3.4	Review Lab	Midterm 1: Chapters 1 & 2
Week 5: June 12-16	Sec. 3.5 <i>Start homework 5</i>	Sec. 3.6 <i>Homework 4 due</i>	Sec 4.1	Lab 4 <i>Lab 3 due</i>	Sec 4.2 <i>Canvas Quiz 4</i>
Week 6: June 19-23	Juneteenth <i>Start homework 6</i>	Sec. 4.3 <i>Homework 5 due</i>	Sec. 4.4	Lab 5 <i>Lab 4 due</i>	Sec 5.1 <i>Canvas Quiz 5</i> Withdrawal deadline
Week 7: June 26-30	Sec. 5.2 <i>Start homework 7</i>	Sec 5.3 <i>Homework 6 due</i>	Sec. 5.4	Review Lab	Midterm 2: Chapters 3 & 4
Week 8: July 3-7	Sec. 5.5 <i>Start homework 8</i>	Independence Day <i>Homework 7 due</i>	Section 5.6	Lab 6 <i>Lab 5 due</i>	Sec 6.1 <i>Canvas quiz 6</i>
Week 9: July 10-14	Sec 6.2 <i>Start homework 9</i>	Sec. 7.1 <i>Homework 8 due</i>	Sec 7.2	Lab 7 <i>Lab 6 due</i>	Sec 7.3 <i>Canvas quiz 7</i>
Week 10: July 17-21	Sec. 7.4 <i>Start homework 10</i>	Sec 7.5 <i>Homework 9 due</i>	Wrap up loose ends for chapters 6 & 7	Review Lab	Midterm 3: Chapters 5-7.3
Week 11: July 24-28	Pioneer Day <i>Start homework 11</i>	Section 10.1 <i>Homework 10 due</i>	Sec. 10.2	Sec. 10.3 <i>Lab 7 due</i>	Sec. 10.4 <i>Canvas quiz 8</i>
Week 12: July 31-Aug 4	Sec 10.5	Review for Final <i>Homework 11 due</i>	Review for Final	Final Exam Comprehensive	

Week by Week Guide:

Week 1: Introduction to differential equations; integrals as general and particular solutions; slope fields and solution curves; separable equations; applications to exponential growth and decay models

Week 2: Linear first-order equations; applications to mixture problems; population models; equilibrium solutions and stability; bifurcations; acceleration-velocity models

Week 3: Numerical approximation: Euler's method and the Improved Euler Method; introduction to linear systems of equations

Week 4: Matrices; Gaussian elimination; reduced row echelon form; matrix operations

Week 5: Inverses of matrices; determinants; vector spaces; the vector space $\mathbb{R}^n$ and subspaces

Week 6: Linear combinations, span, and linear independence; bases and dimension; second-order linear differential equations; superposition; applications to mass-spring systems

Week 7: General solutions to second order linear differential equations; homogeneous equations with constant coefficients; mechanical vibrations

Week 8: Non-homogeneous equations and the method of undetermined coefficients; forced oscillations and resonance; introduction to eigenvalues and eigenvectors

Week 9: Diagonalization of matrices; first order systems of linear differential equations; applications to many oscillating particles and connected tanks of fluids; the eigenvalue method for linear systems

Week 10: Phase plane diagrams for systems of differential equations; mechanical applications

Week 11-12: Laplace Transforms; solving IVPs with transforms; partial fractions and translations; discontinuous forcing; pulse functions; convolutions

Course Learning Outcomes

- Be able to model dynamical systems that arise in science and engineering, by using general principles to derive the governing differential equations or systems of differential equations. These principles include linearization, compartmental analysis, Newton's laws, conservation of energy and Kirchoff's law.
- Learn solution techniques for first order separable and linear differential equations. Solve initial value problems in these cases, with applications to problems in science and engineering. Understand how to approximate solutions even when exact formulas do not exist. Visualize solution graphs and numerical approximations to initial value problems via slope fields.
- Become fluent in matrix algebra techniques, in order to be able to compute the solution space to linear systems and understand its structure; by hand for small problems and with technology for large problems.
- Be able to use the basic concepts of linear algebra such as linear combinations, span, independence, basis and dimension, to understand the solution space to linear equations, linear differential equations, and linear systems of differential equations.
- Understand the natural initial value problems for first order systems of differential equations, and how they encompass the natural initial value problems for higher order differential equations and general systems of differential equations.
- Learn how to solve constant coefficient linear differential equations via superposition, particular solutions, and homogeneous solutions found via characteristic equation analysis. Apply these techniques to understand the solutions to the basic unforced and forced mechanical and electrical oscillation problems.
- Learn how to use Laplace transform techniques to solve linear differential equations, with an emphasis on the initial value problems of mechanical systems, electrical circuits, and related problems.
- Be able to find eigenvalues and eigenvectors for square matrices. Apply these matrix algebra concepts to find the general solution space to first and second order constant coefficient homogeneous linear systems of differential equations, especially those arising from compartmental analysis and mechanical systems.
- Students will also improve problem solving fluency, to read and interpret problem objectives, be able to select and execute appropriate methods to achieve the aforementioned objectives, and be able to interpret and communicate results.

Resources for Math Support

I encourage each of you to utilize my office hours, but the department also offers the following resources:

- **The Math Center:** The Math Center is located in the basement connecting JWB and LCB and offers free drop-in tutoring. For more information, visit <https://www.math.utah.edu/undergraduate/mathcenter.php>.
- **Online Math Lectures:**
Recorded lectures and accompanying notes for a variety of lower division mathematics courses are available at <https://www.math.utah.edu/lectures/index.php>. Though there are no lectures for this course material, these may help you brush up on prerequisite material from previous courses.

Accommodations and Statements

Academic Honesty: Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Students are encouraged to review the Student Code for the University of Utah at <https://regulations.utah.edu/academics/6-400.php>.

COVID-19 Statement: The COVID-19 guidelines for the University of Utah are adapted often due to the ever-changing status of the pandemic. For the most up-to-date information regarding the campus guidelines, visit <https://coronavirus.utah.edu>.

Inclusion and Support

We must treat every individual with respect. We are diverse in many ways, and this diversity is fundamental to building and maintaining an equitable and inclusive campus community. Diversity can refer to multiple ways that we identify ourselves, including but not limited to: race, color, national origin, language, sex, disability, age, sexual orientation, gender identity, religion, creed, ancestry, belief, or veteran status. One thing that unites us is that we all experience stressful and difficult events as a normal part of life. I believe your mental health is an important part of your academic success. **Please feel free to reach out to me about any difficulty you may be having that may impact your performance in this course as soon as it occurs and before it becomes unmanageable.** I strongly encourage you to contact the many other support services on campus that stand ready to assist you. The resources I have listed below are by no means exhaustive.

The Americans with Disabilities Act: The University of Utah seeks to provide equal access to its programs, services and activities for people with disabilities. If you will need accommodations in the class, reasonable prior notice needs to be given to the Center for Disability and Access, 162 Olpin Union Building, 801-581-5020. CDA will work with you and the instructor to make arrangements for accommodations. All written information in this course can be made available in alternative format with prior notification to the Center for Disability and Access.

Addressing Sexual Misconduct: Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a Civil Rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran status or genetic information. If you or someone you know has been harassed or assaulted on the basis of your sex, including sexual orientation or gender identity/expression, you are encouraged to report it to the University's Title IX Coordinator; Director, Office of Equal Opportunity and Affirmative Action, 135 Park Building, 801-581-8365, or to the Office of the Dean of Students, 270 Union Building, 801-581-7066. For support and confidential consultation, contact the Center for Student Wellness, 426 SSB, 801-581-7776. To report to police, contact the Department of Public Safety, 801-585-2677(COPS).

Discrimination and Harassment: If you or someone you know has been harassed or assaulted, you are encouraged to report it to the Title IX Coordinator in the Office of Equal Opportunity and Affirmative Action, 135 Park Building, 801-581-8365, or Office of the Dean of Students, 270 Union Building, 801-581-7066. To report to the police, contact the Department of Public Safety, 801-585-2677(COPS). Please see Student Bill of Rights, section E <http://regulations.utah.edu/academics/6-400.php>. I will listen and believe you if someone is threatening you.

The University Counseling Center: The University Counseling Center (UCC) provides developmental, preventive, and therapeutic services and programs that promote the intellectual, emotional, cultural, and

social development of University of Utah students. They advocate a philosophy of acceptance, compassion, and support for those they serve, as well as for each other. They aspire to respect cultural, individual and role differences as they continually work toward creating a safe and affirming climate for individuals of all ages, cultures, ethnicities, genders, gender identities, languages, mental and physical abilities, national origins, races, religions, sexual orientations, sizes and socioeconomic statuses. More information about the counseling center, including ways to contact them, can be found here: <https://counselingcenter.utah.edu/>.

Wellness Statement: Personal concerns such as stress, anxiety, relationship difficulties, depression, cross-cultural differences, etc., can interfere with a student's ability to succeed and thrive at the University of Utah. For helpful resources contact the Center for Student Wellness at www.wellness.utah.edu or 801-581-7776.